

Directed Kneser graphs with the acyclicity–intersection property exist only for $b \leq 2$ or $k \leq b + 1$

A complete answer to Problem 5.40 of Hochstättler, Schröder and Steiner

github.com/graph-theory-AI

Machine-generated note (see provenance box)

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Provenance

The mathematics in this note was produced without human intervention, in three stages. (1) The classification and its proof were found by the language model `gpt-5.6-sol` in a single autonomous pass on 2026-09-01, as part of a large sweep over open problems catalogued from arXiv (catalog id 1812.02420__03; original writeup in `attacks/1812.02420__03/output.md`). (2) An independent adversarial referee agent (`claude-fable-5`) re-derived every step, confirmed the problem statement and the digraph conventions verbatim against the source paper, verified the three positive constructions exhaustively over all subfamilies (up to $2^{21} - 1$ families for $b = 2, k = 7$), and confirmed the key impossibility $(k, b) = (5, 3)$ by an independent finite computation not relying on the structural lemma; its verdict was **CONFIRMED**. Its report is reproduced verbatim in [Appendix B](#). (3) On 2026-09-09 the writeup was rewritten into the present note by `claude-fable-5.1`: the text was reorganised with full proofs in [Appendix A](#), and three clarifications taken from the referee’s report were added and are marked as such (the reason the closed walk in [Lemma 2.1](#) lies on a proper subset, the second cyclic order in the $b = 2$ triangle, and the remarks on parallel arcs and on $b = 0$). No other mathematical content was changed. **No human mathematician has checked this argument.** We circulate it to obtain exactly that: is the proof correct, and is the result new?

Abstract

Hochstättler, Schröder and Steiner asked (Problem 5.40 of *On the complexity of digraph colourings and vertex arboricity*) whether for all $k \geq b$ there is a digraph $\vec{K}(k, b)$ on the vertex set $\binom{[k]}{b}$ such that the subdigraph induced by a family of b -sets is acyclic if and only if the family has a common element. Such “directed Kneser graphs” would make b -tuple k -colourings of digraphs equivalent to circular homomorphisms. We show that, in the source paper’s convention (loopless digraphs, antiparallel arcs allowed, a digon being a directed cycle), $\vec{K}(k, b)$ exists if and only if $b \leq 2$ or $k \leq b + 1$. In particular no $\vec{K}(5, 3)$ exists, so the answer to Problem 5.40 is negative in general. The negative part rests on a structural lemma: an inclusion-minimal family with empty intersection must induce exactly a directed Hamilton cycle; two overlapping minimal edge covers of K_{b+2} then force incompatible induced graphs.

1 Introduction

Throughout, digraphs are finite and loopless, and antiparallel arcs are allowed; a pair of antiparallel arcs (a *digon*) is a directed cycle of length two, so a digraph containing a digon is not acyclic. These are exactly the conventions of the source paper [1], which states that its digraphs are loopless but may have anti-parallel arcs and calls a digon a directed cycle of length two. For a digraph D and $\mathcal{F} \subseteq V(D)$ we write $D[\mathcal{F}]$ for the induced subdigraph, and $[k] = \{1, \dots, k\}$.

In their study of fractional and b -tuple colourings of digraphs, Hochstättler, Schröder and Steiner pose the following problem.

Problem 1.1 ([1, Problem 5.40]). For given $k, b \in \mathbb{N}$ with $k \geq b$, is there a directed graph $\vec{K}(k, b)$ with vertex set $\binom{[k]}{b}$ such that the following holds? The subdigraph of $\vec{K}(k, b)$ induced by any $\{B_1, \dots, B_l\} \subseteq \binom{[k]}{b}$ is acyclic if and only if $\bigcap_{i=1}^l B_i \neq \emptyset$.

Such directed Kneser graphs would allow b -tuple k -colourings of digraphs to be characterised by circular homomorphisms to $\vec{K}(k, b)$, tying the fractional dichromatic number to circular homomorphism theory. The source paper confirms the case $k = b + 1$: the directed cycle of length $b + 1$ whose vertices are the sets $[b + 1] \setminus \{i\}$ has the required property. We call a digraph D on $\binom{[k]}{b}$ *admissible for* (k, b) if for every nonempty $\mathcal{F} \subseteq \binom{[k]}{b}$,

$$D[\mathcal{F}] \text{ is acyclic} \iff \bigcap_{B \in \mathcal{F}} B \neq \emptyset. \quad (1)$$

All integers $k \geq b$ below are positive.

Theorem 1.2 (Main theorem). *For positive integers $k \geq b$, a digraph admissible for (k, b) exists if and only if*

$$b \leq 2 \quad \text{or} \quad k \leq b + 1.$$

Corollary 1.3. *The answer to Problem 1.1 is negative in general: for every $b \geq 3$ an admissible digraph exists only for $k \in \{b, b + 1\}$. The smallest pair without an admissible digraph is $(k, b) = (5, 3)$.*

Proof. Immediate from Theorem 1.2: for $b \geq 3$ and $k \geq b + 2$ no admissible digraph exists, and $(5, 3)$ is the lexicographically smallest such pair with $k \geq b$. $\square$

Idea of proof. Negative part. If $\mathcal{M}$ is an inclusion-minimal family with empty intersection and $|\mathcal{M}| \geq 3$, then $D[\mathcal{M}]$ must contain a directed cycle, the cycle must use every vertex of $\mathcal{M}$ (a shorter cycle would sit on a proper subfamily, which has nonempty intersection and so must induce an acyclic digraph), and there can be no further arc (a chord or a reversed arc would again close a directed cycle on a proper subfamily). So $D[\mathcal{M}]$ is exactly a directed Hamilton cycle and its underlying graph a chordless cycle (Lemma 2.1). For $k = b + 2$ the b -sets are complements of edges of K_{b+2} and empty intersection means being an edge cover. A full star at a vertex v is a minimal cover with $b + 1 \geq 4$ edges, so it induces a chordless C_{b+1} ; pick two nonadjacent cycle vertices B_{vp}, B_{vq} and replace the edges vp, vq by pq : this is a minimal cover with b edges, inducing a chordless C_b . Removing B_{vp}, B_{vq} from the first cycle leaves $b - 3$ edges, removing B_{pq} from the second leaves a path with $b - 2$ edges, yet both are the same induced graph (Proposition 3.1). Larger k reduce to $k = b + 2$ by restricting to $\binom{U}{b}$ for a $(b + 2)$ -set U . *Positive part.* For $k = b$ a single vertex works and for $k = b + 1$ the directed $(b + 1)$ -cycle; for $b = 1$ the complete bidirected digraph; for $b = 2$ put digons between disjoint pairs and orient intersecting pairs $\{x, a\}, \{x, c\}$ by the cyclic order read from x , so that a common element yields a transitive tournament while an empty intersection yields a digon or a directed triangle (Proposition 4.1). [full proof]

2 Minimal empty-intersection families

Lemma 2.1 (Structural lemma). *Let D be admissible for (k, b) and let $\mathcal{M} \subseteq \binom{[k]}{b}$ be inclusion-minimal subject to $\bigcap_{B \in \mathcal{M}} B = \emptyset$. If $|\mathcal{M}| \geq 3$, then the arcs of $D[\mathcal{M}]$ are exactly the arcs of a directed Hamilton cycle of $\mathcal{M}$. Consequently, in the undirected graph H on $\binom{[k]}{b}$ in which two vertices are adjacent whenever some arc of D joins them, $H[\mathcal{M}]$ is a chordless cycle of length $|\mathcal{M}|$.*

Idea of proof. $D[\mathcal{M}]$ is not acyclic, so it contains a directed cycle C ; every proper subfamily of $\mathcal{M}$ has nonempty intersection and induces an acyclic digraph, so C spans $\mathcal{M}$. An extra arc $X \rightarrow Y$ together with the cycle path from Y to X , or a reversed cycle arc together with its partner, forms a directed cycle on a proper subfamily, again impossible. [full proof]

3 Nonexistence for $b \geq 3$ and $k \geq b + 2$

Proposition 3.1. *If $b \geq 3$ and $k \geq b + 2$, then no digraph is admissible for (k, b) .*

Idea of proof. Reduce to $k = n := b + 2$ by restriction. Identify each b -set with the complementary edge e of K_n , $B_e = [n] \setminus e$; then a family has empty intersection iff the corresponding edges cover $[n]$. The star $\mathcal{S}_v$ at a vertex v and the family $\mathcal{T} = \{vx : x \notin \{v, p, q\}\} \cup \{pq\}$ are inclusion-minimal edge covers of sizes $n - 1$ and $n - 2$; by Lemma 2.1 they induce chordless cycles C_{n-1} and C_{n-2} in H . Choosing B_{vp}, B_{vq} nonadjacent on the first cycle (possible as $n - 1 \geq 4$), the common subfamily $\mathcal{R} = \mathcal{S}_v \setminus \{vp, vq\}$ induces $n - 5$ edges according to the first cycle and $n - 4$ according to the second. [\[full proof\]](#)

Remark 3.2 (The case $(5, 3)$). For $(k, b) = (5, 3)$ the argument reads: the star at 1 is a minimal 4-edge cover of K_5 and must induce a chordless C_4 ; two opposite vertices of that C_4 correspond to edges $1p, 1q$, and replacing them by pq gives a minimal 3-edge cover, which would have to induce a triangle, contradicting the absence of the chord in the C_4 . The referee verified this instance by a second, independent finite argument that does not use Lemma 2.1: after imposing the constraints of (1) on the four vertices of the star family, exactly 6 of the 2^{12} digraphs survive, all directed Hamilton 4-cycles, and each leads to an empty-intersection triple that is forced to be acyclic (Appendix B).

4 Existence for $b \leq 2$ or $k \leq b + 1$

Proposition 4.1. *An admissible digraph exists in each of the following cases.*

- (a) $k = b$: the edgeless digraph on the single vertex $[k]$.
- (b) $k = b + 1$: the directed cycle $B_1 \rightarrow B_2 \rightarrow \dots \rightarrow B_k \rightarrow B_1$ on $B_i = [k] \setminus \{i\}$.
- (c) $b = 1$: the complete bidirected digraph on the singletons $\{i\}$, $i \in [k]$.
- (d) $b = 2$, any $k \geq 2$: fix a cyclic order on $[k]$ and, for $x \in [k]$, let $<_x$ be the linear order on $[k] \setminus \{x\}$ obtained by reading the cyclic order starting immediately after x . Put both arcs between disjoint 2-sets, and for $A = \{x, a\}$, $B = \{x, c\}$ with $a \neq c$ put the single arc $A \rightarrow B$ iff $a <_x c$.

Idea of proof. (a) and (b): $\bigcap_{i \in I} B_i = [k] \setminus I$ is empty only for $I = [k]$, and every proper induced subdigraph of a directed cycle is acyclic. (c): distinct singletons have empty intersection and every pair of vertices carries a digon. (d): a family with common element x is a set of vertices $\{x, a\}$ whose arcs all increase in $<_x$, hence a transitive tournament; a family with empty intersection either contains two disjoint sets (a digon) or is pairwise intersecting without a common element, in which case it contains $\{x, a\}$, $\{x, c\}$, $\{a, c\}$, and these form a directed triangle in either cyclic order of x, a, c . [\[full proof\]](#)

5 Remarks

Remark 5.1 (Conventions). The classification lives in the source paper’s convention that antiparallel arcs are allowed and a digon is a directed cycle. This is not a strawman reading: if digons were forbidden, two disjoint b -sets could never induce a non-acyclic two-vertex digraph, so the problem would fail trivially whenever $k \geq 2b$, and the authors’ own confirmed case $b = 1$, $k = 2$ is the digon. The source paper also allows parallel arcs; the writeup works with simple digraphs (digons allowed). As the referee notes, this is without loss of generality, since arc multiplicities never affect the acyclicity of an induced subdigraph, and Lemma 2.1 is then literally correct (a parallel duplicate of a cycle arc is not excluded by its proof, but only the underlying graph H is used afterwards).

Remark 5.2 (Degenerate parameters). Theorem 1.2 is stated for positive integers. If $\mathbb{N}$ were read as containing 0, the pair $(k, 0)$ would fail trivially: the single vertex $\emptyset$ has empty intersection but a one-vertex loopless digraph is acyclic. For the empty subfamily the usual conventions $\bigcap_{\emptyset} = [k]$ and “the empty digraph is acyclic” agree, so (1) may equivalently be required for all subfamilies.

Remark 5.3 (Computational checks). The referee's scripts (verification/scripts/1812.02420__03/) verified (1) over every nonempty subfamily for the constructions of [Proposition 4.1](#): the directed cycle for $b = 1, \dots, 5$, the bidirected complete digraph for $k \leq 5$, and the $b = 2$ construction for $k = 2, \dots, 7$ (for $k = 7$ all $2^{2^1} - 1 = 2\,097\,151$ families). They also confirmed the edge-cover correspondence and the minimality of $\mathcal{S}_v$ and $\mathcal{T}$ for $n = 5, 6, 7$, and enumerated the inclusion-minimal empty-intersection families of $\binom{[5]}{3}$ (30 of size 3 and 5 of size 4, exactly the minimal edge covers of K_5). No violation was found.

Remark 5.4 (Consequences and novelty). The intended application in [1] was to express b -tuple k -colourings of digraphs as circular homomorphisms to $\vec{K}(k, b)$; [Theorem 1.2](#) shows that this cannot be set up in the proposed form for any $b \geq 3$ and $k \geq b + 2$. The referee's search found no prior construction or refutation of these digraphs; the related papers [2, 3] study dichromatic numbers of Kneser-type digraphs with canonical orientations, not the intersection-acyclicity property. This has not been checked by a human.

A Full proofs

Lemma 2.1 (Structural lemma). *Let D be admissible for (k, b) and let $\mathcal{M} \subseteq \binom{[k]}{b}$ be inclusion-minimal subject to $\bigcap_{B \in \mathcal{M}} B = \emptyset$. If $|\mathcal{M}| \geq 3$, then the arcs of $D[\mathcal{M}]$ are exactly the arcs of a directed Hamilton cycle of $\mathcal{M}$. Consequently, in the undirected graph H on $\binom{[k]}{b}$ in which two vertices are adjacent whenever some arc of D joins them, $H[\mathcal{M}]$ is a chordless cycle of length $|\mathcal{M}|$.*

Proof. Since $\bigcap_{B \in \mathcal{M}} B = \emptyset$, property (1) says that $D[\mathcal{M}]$ is not acyclic, so it contains a directed cycle C (possibly of length two). Let $\mathcal{M}' = V(C)$. If $\mathcal{M}' \subsetneq \mathcal{M}$, then $D[\mathcal{M}']$ contains C and is not acyclic, so by (1) $\bigcap_{B \in \mathcal{M}'} B = \emptyset$, contradicting the inclusion-minimality of $\mathcal{M}$. Hence C uses every vertex of $\mathcal{M}$: it is a directed Hamilton cycle $v_1 \rightarrow v_2 \rightarrow \dots \rightarrow v_m \rightarrow v_1$ of $\mathcal{M}$, $m = |\mathcal{M}| \geq 3$.

We show that $D[\mathcal{M}]$ has no further arc. Suppose $X \rightarrow Y$ is an arc of $D[\mathcal{M}]$ that is not one of the m cycle arcs. Since D is loopless, $X \neq Y$, and since $X \rightarrow Y$ is not a cycle arc, Y is not the successor of X on C . Follow C forward from Y until X and then use $X \rightarrow Y$: this is a directed cycle whose vertex set omits the successor of X on C (and every vertex strictly between X and Y in the forward direction), hence lies on a proper subfamily $\mathcal{M}'' \subsetneq \mathcal{M}$. [Clarification added from the referee's report: this is why the subfamily is proper.] Then $D[\mathcal{M}'']$ is not acyclic while $\bigcap_{B \in \mathcal{M}''} B \neq \emptyset$ by minimality of $\mathcal{M}$, contradicting (1). This covers in particular the reverse $v_{i+1} \rightarrow v_i$ of a cycle arc: it forms with $v_i \rightarrow v_{i+1}$ a digon on the two-element subfamily $\{v_i, v_{i+1}\}$, which is proper because $m \geq 3$.

Hence the arc set of $D[\mathcal{M}]$ is exactly that of C . Two vertices of $\mathcal{M}$ are then adjacent in H iff they are consecutive on C , so $H[\mathcal{M}]$ is a chordless cycle of length m . $\square$

Proposition 3.1. *If $b \geq 3$ and $k \geq b + 2$, then no digraph is admissible for (k, b) .*

Proof. *Reduction to $k = b + 2$.* Suppose D is admissible for (k, b) with $k > b + 2$. Choose $U \subseteq [k]$ with $|U| = b + 2$ and consider $D' = D[\binom{U}{b}]$. For $\mathcal{F} \subseteq \binom{U}{b}$ we have $D'[\mathcal{F}] = D[\mathcal{F}]$ and the intersection of $\mathcal{F}$ is the same whether computed in U or in $[k]$, so D' satisfies (1) for every nonempty $\mathcal{F} \subseteq \binom{U}{b}$. Relabelling U as $[b + 2]$ makes D' admissible for $(b + 2, b)$. It therefore suffices to show that no digraph is admissible for (n, b) with $n = b + 2$ and $b \geq 3$.

Edge covers. Every b -subset of $[n]$ is the complement of a unique 2-subset, i.e. of a unique edge of the complete graph K_n on $[n]$. For an edge e write $B_e = [n] \setminus e$; this is a bijection between $E(K_n)$ and $\binom{[n]}{b}$. For a set $\mathcal{E}$ of edges,

$$\bigcap_{e \in \mathcal{E}} B_e = [n] \setminus \bigcup_{e \in \mathcal{E}} e,$$

so $\{B_e : e \in \mathcal{E}\}$ has empty intersection iff every vertex of $[n]$ lies on some edge of $\mathcal{E}$, i.e. iff $\mathcal{E}$ is an edge cover of K_n . Inclusion-minimal empty-intersection families thus correspond to inclusion-minimal

edge covers. Assume, for a contradiction, that D is admissible for (n, b) and let H be its underlying graph as in Lemma 2.1; we identify a vertex B_e of H with the edge e .

The star. Fix $v \in [n]$ and let $\mathcal{S}_v = \{vx : x \in [n] \setminus \{v\}\}$, the set of $n - 1$ edges at v . It is an edge cover (v is covered by any of its edges, and $x \neq v$ by vx), and it is inclusion-minimal, since deleting vx leaves x uncovered. Its size is $n - 1 = b + 1 \geq 4 \geq 3$, so Lemma 2.1 applies: $H[\mathcal{S}_v]$ is a chordless cycle C_{n-1} . As $n - 1 \geq 4$, this cycle has two nonadjacent vertices; call them B_{vp} and B_{vq} , with $p, q \in [n] \setminus \{v\}$ distinct.

The modified cover. Let $\mathcal{T} = \{vx : x \in [n] \setminus \{v, p, q\}\} \cup \{pq\}$. It has $(n - 3) + 1 = n - 2$ edges. It is an edge cover: p and q are covered by pq , every $x \notin \{v, p, q\}$ by vx , and v by any edge vx with $x \notin \{v, p, q\}$, which exists because $n - 3 = b - 1 \geq 2$. It is inclusion-minimal: deleting vx leaves x uncovered, and deleting pq leaves both p and q uncovered. Its size is $n - 2 = b \geq 3$, so by Lemma 2.1 $H[\mathcal{T}]$ is a chordless cycle C_{n-2} .

The clash. Let $\mathcal{R} = \mathcal{S}_v \setminus \{vp, vq\} = \{vx : x \notin \{v, p, q\}\} = \mathcal{T} \setminus \{pq\}$, a set of $n - 3$ vertices of H contained in both $\mathcal{S}_v$ and $\mathcal{T}$. Since induced subgraphs of induced subgraphs are induced subgraphs, $H[\mathcal{R}]$ is obtained both from $H[\mathcal{S}_v]$ by deleting B_{vp}, B_{vq} and from $H[\mathcal{T}]$ by deleting B_{pq} .

1. $H[\mathcal{S}_v] \cong C_{n-1}$ and B_{vp}, B_{vq} are nonadjacent on it. Deleting two nonadjacent vertices of a cycle removes four distinct edges, so $H[\mathcal{R}]$ has $(n - 1) - 4 = n - 5$ edges.
2. $H[\mathcal{T}] \cong C_{n-2}$. Deleting one vertex of a cycle leaves a path on $n - 3$ vertices, so $H[\mathcal{R}]$ has $(n - 2) - 2 = n - 4$ edges.

The same graph cannot have both $n - 5$ and $n - 4$ edges. This contradiction shows that no digraph is admissible for $(b + 2, b)$ when $b \geq 3$, and by the reduction none for (k, b) with $k \geq b + 2, b \geq 3$. $\square$

Proposition 4.1. *An admissible digraph exists in each of the following cases.*

- (a) $k = b$: the edgeless digraph on the single vertex $[k]$.
- (b) $k = b + 1$: the directed cycle $B_1 \rightarrow B_2 \rightarrow \dots \rightarrow B_k \rightarrow B_1$ on $B_i = [k] \setminus \{i\}$.
- (c) $b = 1$: the complete bidirected digraph on the singletons $\{i\}, i \in [k]$.
- (d) $b = 2$, any $k \geq 2$: fix a cyclic order on $[k]$ and, for $x \in [k]$, let $<_x$ be the linear order on $[k] \setminus \{x\}$ obtained by reading the cyclic order starting immediately after x . Put both arcs between disjoint 2-sets, and for $A = \{x, a\}, B = \{x, c\}$ with $a \neq c$ put the single arc $A \rightarrow B$ iff $a <_x c$.

Proof. In each case we verify (1) for every nonempty family $\mathcal{F}$ of vertices.

(a) $k = b$. The only vertex is $[k]$, and the only nonempty family is $\{[k]\}$, whose intersection $[k]$ is nonempty since $k = b \geq 1$. A single loopless vertex is acyclic. So (1) holds.

(b) $k = b + 1$. The vertices are $B_i = [k] \setminus \{i\}, i \in [k]$, and for $\emptyset \neq I \subseteq [k]$,

$$\bigcap_{i \in I} B_i = [k] \setminus I,$$

which is empty iff $I = [k]$. The digraph is the directed cycle $B_1 \rightarrow B_2 \rightarrow \dots \rightarrow B_k \rightarrow B_1$. Its only directed cycle is the whole cycle, so $D[\{B_i : i \in I\}]$ is acyclic iff $I \neq [k]$ iff the intersection is nonempty. (For $k = 2$ the cycle is the digon $B_1 \rightleftarrows B_2$, which is consistent with the conventions.)

(c) $b = 1$. The vertices are the singletons $\{i\}, i \in [k]$. Two distinct singletons are disjoint, so a family of singletons has nonempty intersection iff it consists of a single vertex. In the complete bidirected digraph every induced subdigraph on at least two vertices contains a digon, hence is not acyclic, while a single vertex is acyclic. So (1) holds.

(d) $b = 2$. The digraph is well defined: two distinct intersecting 2-sets $A \neq B$ have exactly one common element x , so the rule assigns exactly one arc to the pair $\{A, B\}$ (namely from the set whose other element is $<_x$ -smaller), and disjoint pairs receive a digon.

First let $\mathcal{F}$ have a common element x . Then every member of $\mathcal{F}$ has the form $\{x, a\}$ with $a \in [k] \setminus \{x\}$, distinct members have distinct a , and for two members $\{x, a\}, \{x, c\}$ the unique arc between them goes from $\{x, a\}$ to $\{x, c\}$ iff $a <_x c$. Thus $D[\mathcal{F}]$ is the transitive tournament of the linear order $<_x$ restricted to $\{a : \{x, a\} \in \mathcal{F}\}$, which is acyclic (a directed cycle would contain an arc decreasing in $<_x$).

Now let $\mathcal{F}$ have empty intersection. If $\mathcal{F}$ contains two disjoint members A, B , then $D[\mathcal{F}]$ contains the digon $A \rightleftharpoons B$ and is not acyclic. Otherwise $\mathcal{F}$ is pairwise intersecting. It has at least two members (a single member has nonempty intersection), say $\{x, a\}$ and $\{x, c\}$ with common element x and $a \neq c$. Since x is not a common element of $\mathcal{F}$, some member $F \in \mathcal{F}$ does not contain x ; as F meets $\{x, a\}$ and $\{x, c\}$, it contains a and c , so $F = \{a, c\}$. Thus $\{x, a\}, \{x, c\}, \{a, c\} \in \mathcal{F}$. We show these three vertices induce a directed triangle. The elements x, a, c are distinct; read the cyclic order starting from x .

- If a comes before c , i.e. $a <_x c$, then $\{x, a\} \rightarrow \{x, c\}$. Reading from c , the next elements among $\{x, a\}$ are x then a , so $x <_c a$ and $\{x, c\} \rightarrow \{a, c\}$. Reading from a , c comes before x , so $c <_a x$ and $\{a, c\} \rightarrow \{x, a\}$. Hence $\{x, a\} \rightarrow \{x, c\} \rightarrow \{a, c\} \rightarrow \{x, a\}$ is a directed 3-cycle.
- If c comes before a , exchanging the roles of a and c in the previous item gives the directed 3-cycle $\{x, c\} \rightarrow \{x, a\} \rightarrow \{a, c\} \rightarrow \{x, c\}$. [Clarification added from the referee's report, which checked both cyclic orders; the writeup treats the first.]

In all cases $D[\mathcal{F}]$ is not acyclic. Hence (1) holds for every $k \geq 2$ when $b = 2$. $\square$

Theorem 1.2 (Main theorem). *For positive integers $k \geq b$, a digraph admissible for (k, b) exists if and only if*

$$b \leq 2 \quad \text{or} \quad k \leq b + 1.$$

Proof. Let $k \geq b \geq 1$. If $b \leq 2$, an admissible digraph exists by Proposition 4.1(c) for $b = 1$ and by Proposition 4.1(d) for $b = 2$ (note $k \geq b = 2$). If $k \leq b + 1$, one exists by Proposition 4.1(a) for $k = b$ and Proposition 4.1(b) for $k = b + 1$. In the remaining case $b \geq 3$ and $k \geq b + 2$, Proposition 3.1 shows that none exists. The three cases are exhaustive, which proves the equivalence. $\square$

B Report of the LLM referee (verbatim)

The following is the unedited report of the adversarial referee agent (claude-fable-5, 2026-09-02) on the *original* writeup. Its step numbers refer to that writeup, not to this note. The scripts it mentions are in `verification/scripts/1812.02420__03/`. Treat it as machine output, not as an authority.

B.1 Interpretation

Problem 5.40 of arXiv:1812.02420 (Hochstättler, Schröder, Steiner) asks, for $k \geq b$, whether there is a digraph $\vec{K}(k, b)$ on vertex set $\binom{[k]}{b}$ such that the induced subdigraph on any subfamily is acyclic iff the subfamily has nonempty intersection. The writeup answers with a complete classification: such a digraph exists iff $b \leq 2$ or $k \leq b + 1$; in particular $(k, b) = (5, 3)$ is impossible, so the answer to the open problem is “no” in general.

The only convention on which the verdict could hinge is whether digons (antiparallel arc pairs) are allowed and count as directed cycles. I fetched the source paper (<https://arxiv.org/html/1812.02420>): it states explicitly that digraphs are “considered loopless, but are allowed to have multiple parallel and anti-parallel edges/arcs between vertices” and that a “directed cycle of length two (a digon) is a pair of anti-parallel arcs”. The writeup’s stated convention (“loopless, antiparallel arcs allowed, a directed 2-cycle counts as a directed cycle”) is therefore *exactly* the source paper’s convention, not a strawman. Indeed the problem is only nontrivially open under this convention: any two disjoint b -sets must induce a non-acyclic 2-vertex digraph, which forces digons whenever $k \geq 2b$ — so the authors plainly intended digons. Note also that the paper’s own confirmed case for $b = 1, k = 2$ would be the digon C_2 .

The problem asks a yes/no question for given (k, b) ; the writeup resolves it for every pair of positive integers, negatively in general. A reasonable author of the source paper would consider Problem 5.40 resolved (in the negative — the hoped-for tie between fractional dichromatic number and circular

homomorphisms via these digraphs cannot be set up this way for $b \geq 3, k \geq b + 2$). Interpretation is fair. The writeup restricts to positive integers, which matches the intent (see Caveats for the degenerate $b = 0$ reading).

B.2 Step-by-step findings

step	label	note
1. Theorem statement (exists iff $b \leq 2$ or $k \leq b + 1$)	VALID	Follows from steps 2–17; consistent with the paper’s confirmed case $k = b + 1$.
2. Lemma: inclusion-minimal empty-intersection $\mathcal{M}$, $ \mathcal{M} \geq 3 \Rightarrow D[\mathcal{M}]$ is exactly a directed Hamilton cycle	VALID	See 3–4. Computationally corroborated: for $(5, 3)$, the only 4-vertex digraphs meeting the subfamily constraints on $\mathcal{S}_1$ are the 6 directed Hamilton 4-cycles (Stage A of <code>check_53_impossible.py</code>).
3. Lemma proof: the cycle C spans $\mathcal{M}$	VALID	If $V(C) \subsetneq \mathcal{M}$, then $D[V(C)]$ non-acyclic forces $\bigcap_{V(C)} B = \emptyset$, contradicting inclusion-minimality. Correct.
4. Lemma proof: no arcs besides the Hamilton cycle	VALID	A chord $X \rightarrow Y$ closes a directed cycle on a proper subset (the cycle path $Y \rightarrow \dots \rightarrow X$ omits at least one vertex strictly between X and Y since $X \rightarrow Y$ is not a cycle arc); a reversed cycle arc gives a digon on 2 vertices, proper since $ \mathcal{M} \geq 3$. Both contradict minimality. (Micro-remark: a <i>parallel duplicate</i> of a cycle arc is not excluded by this argument, but the writeup’s own convention has no parallel arcs, restricting to simple digraphs is WLOG for this problem, and only the underlying graph H is used downstream — harmless.)
5. §2 correspondence: $B_e = [n] \setminus e$, empty intersection $\Leftrightarrow \mathcal{E}$ is an edge cover of K_n ($n = b + 2$)	VALID	$\bigcap B_e = [n] \setminus \bigcup e$. Re-derived by hand and verified by code (exhaustively for $n = 5$, sampled for $n = 6, 7$).
6. $\mathcal{S}_v$ is an inclusion-minimal edge cover of size $n - 1$, so $H[\mathcal{S}_v] \cong$ chordless C_{n-1}	VALID	Deleting vx uncovers x ; minimality verified by code for $n = 5, 6, 7$. Lemma applies since $n - 1 = b + 1 \geq 4 \geq 3$; the induced underlying graph of a directed Hamilton cycle on ≥ 3 vertices is a chordless cycle.

step	label	note
7. Existence of nonadjacent B_{vp}, B_{vq} in C_{n-1}	VALID	Needs $n - 1 \geq 4$, i.e. $b \geq 3$ – exactly the hypothesis.
8. $\mathcal{T} = \{vx : x \notin \{v, p, q\}\} \cup \{pq\}$ is an inclusion-minimal edge cover of size $n - 2$, so $H[\mathcal{T}] \cong$ chordless C_{n-2}	VALID	Cover and minimality re-derived (v is covered since $n - 3 \geq 2$) and verified by code for $n = 5, 6, 7$. Lemma applies since $n - 2 = b \geq 3$.
9. Edge-count clash on $\mathcal{R} = \mathcal{S}_v \setminus \{vp, vq\}$: $n - 5$ vs $n - 4$ edges	VALID	Deleting 2 nonadjacent vertices from an induced chordless C_{n-1} leaves $n - 5$ edges; deleting B_{pq} from the induced chordless C_{n-2} leaves a path with $n - 4$ edges. Same induced graph $H[\mathcal{R}]$, different counts – contradiction. Arithmetic re-derived and checked for $n = 5, \dots, 7$. Note induced-of-induced equals induced, so both computations describe the same graph. Nonexistence for $k = b + 2$, $b \geq 3$ follows.
10. Extension to all $k \geq b + 2$ by restriction to $\binom{U}{b}$, $ U = b + 2$	VALID	The iff-property is inherited by any induced sub-collection; relabel $U \rightarrow [b + 2]$.
11. $k = b$: one-vertex edgeless digraph	VALID	The only nonempty family is $\{[k]\}$, intersection $[k] \neq \emptyset$ ($b \geq 1$), one loopless vertex is acyclic.
12. $k = b + 1$: directed $(b + 1)$ -cycle on $B_i = [k] \setminus \{i\}$	VALID	$\bigcap_{i \in I} B_i = [k] \setminus I$, empty iff $I = [k]$; the full cycle is the unique cyclic induced subdigraph. Exhaustively verified by code for $b = 1, \dots, 5$. Matches the source paper's own example.
13. $b = 1$: complete bidirected digraph	VALID	Distinct singletons intersect iff the family is a single vertex; every 2-set of vertices carries a digon. Exhaustively verified for $k \leq 5$.
14. $b = 2$ construction well defined	VALID	Two distinct intersecting 2-sets have a unique common element, so the $<_x$ rule assigns exactly one arc per intersecting pair; disjoint pairs get digons.
15. $b = 2$: common element $\Rightarrow$ acyclic	VALID	All members contain x ; the arcs form the transitive tournament of the linear order $<_x$.

step	label	note
16. $b = 2$: empty intersection $\Rightarrow$ digon or triangle present	VALID	Either two disjoint members (digon), or pairwise-intersecting with no common element: two members meet in some x , a member avoiding x must be $\{a, c\}$ — the classical Helly-type fact for 2-sets, correctly derived.
17. $b = 2$: the triangle $\{x, a\}, \{x, c\}, \{a, c\}$ is a directed 3-cycle	VALID	I checked both cyclic orders: order x, a, c gives $\{x, a\} \rightarrow \{x, c\} \rightarrow \{a, c\} \rightarrow \{x, a\}$; order x, c, a gives the reverse 3-cycle. Either way non-acyclic. Exhaustively verified by code for $k = 2, \dots, 7$ (all $2^{\binom{k}{2}} - 1$ families).
18. Conclusion / boxed classification	VALID	Cases $b \leq 2, k \leq b + 1$, and $b \geq 3 \wedge k \geq b + 2$ are exhaustive and consistent.

No step is a GAP or ERROR.

B.3 Reference check

The writeup invokes no external theorems. The only reference is the source paper itself, which I fetched (<https://arxiv.org/html/1812.02420>, journal version DMTCS vol. 22 no. 1, 2020):

- **Problem statement:** the paper’s concluding open problem asks exactly the quoted question (digraph on $\binom{[k]}{b}$, induced subfamily acyclic iff nonempty intersection). Confirmed verbatim.
- $k = b + 1$ **case:** the paper states “whenever $k = b + 1$, such a digraph is given by the directed cycle of length $b + 1$ whose vertices are associated with the sets $\{1, \dots, b + 1\} \setminus \{i\}$ ” — matching the writeup’s §3.1. Confirmed.
- **Digraph conventions:** the paper explicitly allows anti-parallel arcs and defines a digon as a directed cycle of length two. The writeup’s load-bearing convention is the paper’s own. Confirmed.

Prior-work search (WebSearch, plus the catalog’s own auto-review of 2026-05-15): no paper constructing or refuting these directed Kneser graphs was found; the related papers arXiv:2309.16565 and arXiv:2511.12553 study dichromatic numbers of Kneser-type digraphs but not the intersection-acyclicity property. I could not find this classification in the literature, so ALREADY_KNOWN does not apply on present evidence.

B.4 Computational check

Scripts in `verification/scripts/1812.02420__03/` (pure Python 3, no external packages; total runtime under 10 seconds):

1. `check_53_impossible.py` — an *independent* finite proof of the key negative case $(5, 3)$, not relying on the writeup’s lemma. It verifies the set-level facts (every two 3-subsets of $[5]$ intersect, so no digon may appear anywhere; the family $\mathcal{S}_1 = \{[5] \setminus \{1, x\}\}$ has empty intersection but all its triples intersect; each $\{V_x, V_y, U_{xy}\}$ with $U_{xy} = \{1, x, y\}$ has empty intersection with

nonempty pairwise intersections). It then enumerates **all** $2^{12} = 4096$ **digraphs** on the 4 vertices of $\mathcal{S}_1$ subject to the subfamily constraints: exactly **6 survive, all directed Hamilton 4-cycles** (independently confirming the lemma’s prediction), each with an arcless pair; and enumerates all 2^4 arc-configurations on $\{V_x, V_y, U_{xy}\}$ given no $V_x - V_y$ arcs and no digons: **0 of them contain a directed cycle**, so the empty-intersection family $\{V_x, V_y, U_{xy}\}$ would be acyclic — contradiction. Hence no $\vec{K}(5, 3)$ exists. It also enumerates all inclusion-minimal empty-intersection families of $\binom{[5]}{3}$: **30 of size 3 and 5 of size 4**, exactly the minimal edge covers of K_5 , as the writeup’s §2 correspondence implies.

2. `check_constructions.py` — exhaustive verification of the iff-property over *every* nonempty subfamily for: the directed cycle for $k = b + 1$, $b = 1, \dots, 5$; the complete bidirected digraph for $b = 1$, $k \leq 5$; and the writeup’s $b = 2$ cyclic-order construction for $k = 2, \dots, 6$ (up to $2^{15} - 1 = 32767$ families). **Zero violations.**
3. `check_b2_k7.py` — bitmask-based exhaustive check of the $b = 2$ construction for $k = 7$: all $2^{21} - 1 = 2\,097\,151$ nonempty families. **Zero violations.**
4. `check_section2_covers.py` — verifies for $n = 5, 6, 7$ the §2 correspondence (complement bijection; empty intersection $\Leftrightarrow$ edge cover — exhaustive for $n = 5$, 20000 random families for $n = 6, 7$), that $\mathcal{S}_v$ and $\mathcal{T}$ are inclusion-minimal edge covers of sizes $n - 1$ and $n - 2$, and the edge-count clash $n - 5 \neq n - 4$ (0 vs 1, 1 vs 2, 2 vs 3).

All claimed properties passed; the crucial finite counterexample case $(5, 3)$ is confirmed impossible by a computation independent of the writeup’s argument.

B.5 Caveats

- **Digon convention (declared by the writeup).** The whole classification lives in the source paper’s convention (digons allowed and cyclic). Under a digon-free convention the problem is trivially false for all $k \geq 2b$ (and for $b = 1, k \geq 2$), so no reasonable reading changes the negative answer; the writeup flags this correctly in its caveats field.
- $b = 0$. If $\mathbb{N}$ were read as containing 0, the pair $(k, 0)$ fails trivially (the single vertex $\emptyset$ gives an empty intersection but a 1-vertex loopless digraph is acyclic), and the writeup’s “ $k = b$ works” would be wrong for $b = 0$. The writeup explicitly restricts to positive integers, which is clearly the intended reading; the classification statement should be read with $b \geq 1$.
- **Empty subfamily.** For $l = 0$ the standard convention $\bigcap_{\emptyset} = [k]$ and the acyclic empty digraph agree; no issue, but the writeup does not discuss it.
- **Multi-arcs.** The paper allows parallel arcs; the writeup silently works with simple digraphs (digons allowed). This is WLOG since arc multiplicities never affect acyclicity of induced subdigraphs, and the lemma’s “exactly a directed Hamilton cycle” is then literally correct.
- **Priority.** My web search and the catalog’s 2026-05 auto-review found no prior resolution, but a negative literature search is never conclusive; the writeup itself flags this. If a prior source surfaces, the verdict would shift to `ALREADY_KNOWN`, not to any error category.
- The claim “ $(5, 3)$ is a counterexample to general existence” is accurate and it is the smallest impossible pair.

B.6 Referee summary

I approached this expecting the usual single-pass failure modes and found none. The interpretation is exactly the source paper’s (I verified the paper’s own digon convention and its Problem statement verbatim). The structural lemma (minimal empty-intersection families induce exactly directed Hamilton cycles) is correct, including the chord/reverse-arc exclusion; the $k = b + 2$ impossibility via the two overlapping minimal edge covers of K_n ($\mathcal{S}_v$ forcing a chordless C_{n-1} , $\mathcal{T}$ forcing a chordless C_{n-2} , clashing on $\mathcal{R}$ with $n - 5 \neq n - 4$ edges) is airtight for all $b \geq 3$, and restriction handles $k > b + 2$. All three positive constructions are correct; I exhaustively machine-verified them (including all 2,097,151

subfamilies for $b = 2, k = 7$), and I machine-verified the $(5, 3)$ impossibility by an independent finite argument that does not use the writeup's lemma. No prior resolution was found in the literature. Verdict: CONFIRMED.

References

- [1] W. Hochstättler, F. Schröder and R. Steiner, On the complexity of digraph colourings and vertex arboricity, *Discrete Math. Theor. Comput. Sci.* 22:1 (2020); arXiv:1812.02420.
- [2] Colouring complete multipartite and Kneser-type digraphs, arXiv:2309.16565 (2023).
- [3] On the dichromatic number of generalized Kneser and Johnson digraphs, arXiv:2511.12553 (2025).